A positive fold test from Nipissing diabase

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Positive fold tests are reported on two remanence components resolved from the same Nipissing diabase sills. These tests conclusively indicate that the region south of the Sudbury Basin in the Southern Structural Province has been subjected to two distinct deformational episodes subsequent to the intrusion of the Nipissing diabase. One event precedes (1850–2150 Ma) and one event postdates (1250–1850 Ma) intrusion of the Sudbury irruptive. Some evidence also exists for a third event that may be related to the development of the Sudbury structure, around 1850 Ma. Evidence that the Nipissing was intruded into already folded Huronian sediments confirms previous suggestions of four separate orogenic pulses in the period 2300–1250 Ma.

On rapporte des tests positifs de plissement sur deux composantes de rémanence résolues pour les mêmes filons-couches de diabase de Nipissing. Ces tests indiquent de façon concluante que la région au sud du bassin de Sudbury dans la province structurale du Sud a été probablement soumise à deux épisodes distincts de déformation. Un épisode précède (1850–2150 Ma) et un épisode suit (1250–1850 Ma) l'intrusion du massif de Sudbury. On possède aussi des preuves pour un troisième épisode qui pourrait être relié au développement de la structure de Sudbury il y a environ 1850 Ma.

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Introduction

A number of authors (Robertson 1964; Church 1968; Card and Pattison 1973) have suggested that the Nipissing diabase intrusions of the Southern Structural Province were emplaced during the later stage of an early major folding of the Huronian sequence, but before secondary deformational events and regional metamorphism. This major deformation of the Nipissing has hitherto not been detected paleomagnetically. Rather, the evidence has tended to support the notion that the magma was intruded into already deformed Huronian sediments. Symons (1970, 1971) and Symons and Londry (1975), for example, on observing small increases of Fisher precision upon tilt correction suggested that only "minor post-consolidation contraction" was permitted by the data. Geochemical evidence also appears to support an original gentle basin and arch configuration (Hriskevich 1968).

For the most part the Huronian rocks into which the Nipissing diabase was intruded have fairly gentle dips (Fig. 1). The western sector of the Southern Structural Province has the overall structure of a block-faulted homocline with dips rarely as great as 45°. The Cobalt plate is characterized by northerly trending basins and domes whose limbs generally dip less than 25° (Roscoe 1969). Only in the central sector between the Elliot Lake and Cobalt plate areas are the Huronian rocks more tightly folded. Dips of beds are steeper than 45° in most places. Along the southern margin of the Sudbury irruptive the beds are locally overturned. No paleomagnetic investigations of the Nipissing have been reported from this more deformed zone. Yet this is the area that Card and Pattison (1973) and Church (1968) refer to when describing the evidence for syntectonic intrusion of the Nipissing.

While no evidence has been forthcoming on the tectonic history of the Nipissing diabase, paleomagnetic investigations have revealed much about the intrusive history of this unit. In particular, it has been shown by positive contact tests of remanence age that there have been two quite distinct phases of Nipissing intrusion, N1 and N2 (Morris 1979). N1 is characterized by remanence directions around $D = 270°$, $I = 65°$, whereas N2 is usually associated with directions around $D = 040°$, $I = -40°$.

It is the purpose of this paper to report a positive fold test from sites sampled in the Nipissing diabase from the region between the Sudbury Basin and the Grenville Front (Fig. 1). Other remanence components resolved from these sites indicate that this zone has been subjected to at least two deformational events, one preceding and one postdating intrusion of the oldest noritic phase of the Sudbury irruptive. Tentative support for the concept of penecontemporaneous intrusion and deformation is provided by the comparison of data from the margins and centres of two large sills.

Paleomagnetic results

Up to seven cores were collected at each of 11 localities in the region between the Grenville Front and the Sudbury Basin (Fig. 1). Each core provided two specimens. Each specimen was subjected to stepwise

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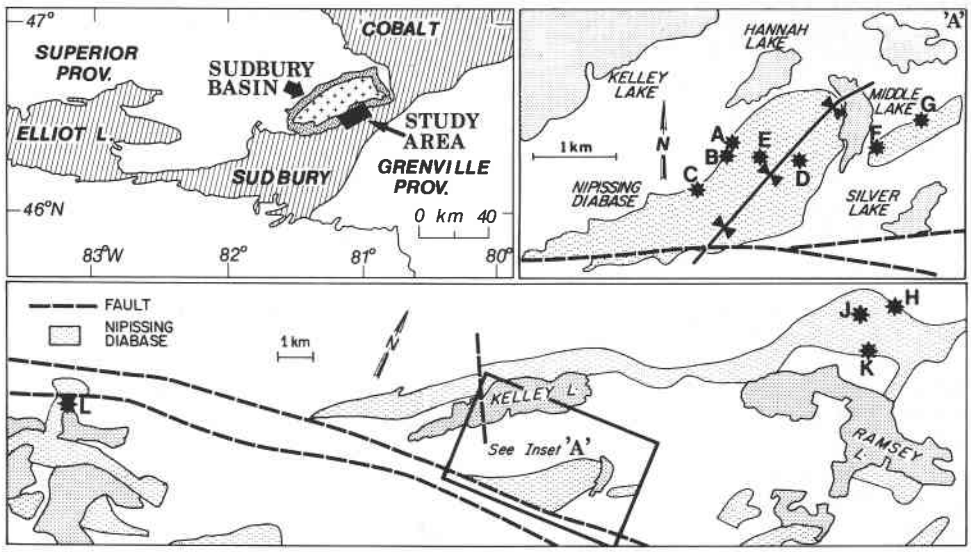


FIG. 1. Diagrammatic map of the sampling localities around the town of Sudbury. For clarity only the distribution of the Nipissing diabase and its relationship to well-known lakes in the area have been shown. For more details on the distribution of pre-Nipissing strata and their structure see the Ontario Department of Mines Map 2017.

cleaning, either by thermal or alternating field (AF) methods. Occasionally, two-stage cleaning (AF followed by thermal) was used. No bulk cleaning was employed. Characteristic remanence directions were identified from the demagnetization trajectories of each specimen by fitting lines and (or) planes to demagnetization segments. Minimum acceptance criteria were: (a) a minimum of four points per line or plane; (b) a minimum angular error of 7.5°. Nearly all specimens retained more than one remanence direction (Table 1). Isolation of individual directions was possible because of the presence of phases with different remanence characteristics.

Figures 2 and 3 summarize the demagnetization characteristics of the various remanence components found in this collection. The west-northwest direction shown in Fig. 3 *a* and *b* is probably the least commonly found direction. As in these examples it invariably has low to moderate unblocking temperatures (<500°C) and low coercive forces (<20 mT). In other specimens it is also found to exist with the opposite polarity. In Fig. 3 *a* and *b* it exhibits simple linear paths on orthogonal demagnetization plots. Directions similar to this have been found in the Sudbury irruptive (N4) where they are often associated with mineralization (Morris 1980). Here these directions will not be considered further.

A more common direction is the shallow positive southeasterly *A* direction (Figs. 2 *a*, *b*, 3*d*). Upon thermal demagnetization this direction generally unblocks in the interval 250–400°C. At site L, where BSY 140 AC was collected, all specimens show similar characteristics: little direction change up to 400°C, and no meaningful direction with further demagnetization. Yet this latter characteristic is often associated with a less pronounced but clearly defined Curie point between 530 and 550°C. Figure 3*d* shows a similar property. The intensity plot shows that a large portion of the *A* remanence is removed after 325°C, but the vector diagram shows that *A* is still present up to 540°C. During the interval 400–540°C both the *A* and *B* directions are being demagnetized, but it is the *A* component that is preferentially being destroyed. AF demagnetization (Fig 2*b*) shows that the *A* direction resides in some mineral with a substantial resistive force. The direction is only unblocked after 50 mT.

The *B* direction (tilt-corrected, it is oriented shallow positive to the southwest; Figs. 2 *a* and *b*, 3 *a*, *b*, and *d*) which is extremely stable under thermal treatment, is carried by two remanence phases. One phase with unblocking temperatures in the range 550–570°C (Figs. 2*a*, 3*b*) is probably low-titanium magnetite. In other specimens this direction shows only minor unblocking up to 620°C (Fig. 3*d*). Usually this high Curie temperature is associated with an extremely stable remanence direction upon demagnetization. This phase of *B* remanence seems to be somewhat resistant to later alteration. The identity of this remanence carrier is uncertain. It is known to coexist in some specimens with the 570°C *B* remanence phase. Possible candidates for remanence carrier are NiFe alloys, aluminum-doped maghemite, or hematite. The extremely high specific magnetization, 3 A/m, could argue against hematite but at the moment it is impossible to decide.

Figure 3*c* presents the special case of specimens

FIG. 2. Demagnetization characteristics of typical specimens from localities in the Nipissing diabase. All are equal-angle stereographic projections showing the tilt-corrected remanence directions obtained upon progressive demagnetization. Crosses (open squares) indicate positively (negatively) inclined remanence vectors. Intensity versus treatment plots show the total intensity remaining after each demagnetization step. Orthogonal plots are of the type used by Roy and Park (1974). (*a*) Thermal demagnetization of a typical specimen BSX 135 BC from the north limb of the syncline. The intensity demagnetization plot shows two well-defined unblocking temperatures (T_{UB}) around 300 and 560°C. The first T_{UB} point coincides with removal of the *A* magnetization and the second T_{UB} point represents decay of the magnetite-residing *B* magnetization. (*b*) AF demagnetization of a specimen BSX 149 BO from the north limb of the syncline. After the initial treatment the direction gradually progresses from a positive southeasterly direction (*A*) to a shallow southerly direction (*B*).

collected from the centre of the main sill. Two directional groups are found. Up to 400°C, the specimen shows an *in situ* direction that is similar to the high temperature / high coercive force tilt-corrected direction found at the other sites in this sill (Fig. 3 *a*, *b*, and *d*). Upon further treatment the directions swing up to a steeply positive westerly direction, which is retained until 575°C. This *in situ* westerly direction after tilt correction is identical to the other tilt-corrected *B* directions shown in Fig. 3 *a*, *b*, and *d*. (Note: this feature has been discovered again at another site at the centre of the same sill some 8 km away, sampled subsequent to the completion of this study.)

TABLE 1. Nipissing diabase

	A magnetization									*B* magnetization								
Site	TR	N_{sp}	N_o	R	κ	α_{95}	$D°$ (*in situ*)	$I°$	$D°$ (corrected)	$I°$	TR	N_{sp}	N_o	R	κ	α_{95}	$D°$ (*in situ*)	$I°$	$D°$ (corrected)	$I°$
A	a, t	7	7	6.903	62	8	87	74	147	33	a, t	5	5	4.936	80	9	235	73	185	35
B	a, t, 2	11	11	10.731	37	8	109	72	147	26	a, t	12	12	11.888	98	4	262	73	188	39
C	a, t, 2	6	6	5.958	118	6	112	73	154	24	a, t, 2	16	16	15.564	35	5	250	55	208	21
E	No reliable determinations																			
F											a, t, 2	8	8	7.904	73	7	35	12	44	−27
H	t	8	8	7.613	18	13	295	−32	304	07	a, t	8	11	7.914	82	6	258	68	192	34
J_1											t	3	3	2.891	19	29	272	72	187	39
K	t	8	8	7.733	26	11	101	31	114	01	a, t	11	11	10.834	60	6	242	75	182	31
D	a, t	3	3	2.982	111	12	151	56	302	60	a, t	5	5	4.952	83	8	184	61	293	33
G	a, t	5	5	4.955	96	8	152	51	349	54	a, t	8	8	7.810	37	9	65	85	345	15
L	a, t, 2	5	5	4.927	55	10	143	55	145	01										
J_2											a, t, 2	8	8	7.660	21	13	191	30	188	−20

NOTES: TR = treatment; a = alternating field; t = thermal; 2 = alternating field followed by thermal; N_{sp} = number of observations of a particular remanence phase during demagnetization. N_o may be greater than N_{sp}. R, κ, α_{95} are the standard Fisher paleomagnetic statistic parameters.

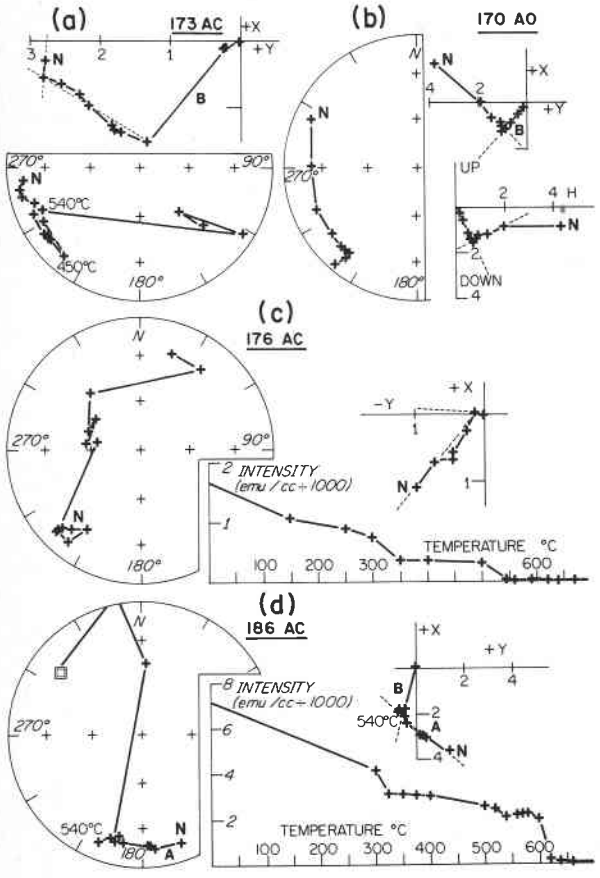


FIG. 3. Demagnetization characteristics of typical specimens from three different levels in the main Sudbury sill. All are equal-angle stereographic projections showing the tilt-corrected (except (*c*), which is *in situ*) remanence directions obtained upon progressive demagnetization. Crosses (open squares) indicate positively (negatively) inclined remanence vectors. Intensity versus treatment plots show the total intensity remaining after each demagnetization step. The format of the vector diagram is of the Roy and Park (1974) type, that is, X versus Y, and Z versus H (horizontal component). (*a*) Thermal demagnetization of a specimen 173 AC from the base of the sill. The stereonet shows a gradual migration from the northwest to the southwest. The vector diagram shows that this trajectory contains three components: an extremely low-temperature natural remanent magnetization (NRM); a second with shallowly inclined negative west-northwesterly (reverse Grenville overprint) direction; and a third higher temperature southwest positive direction (B). (*b*) AF demagnetization of a specimen 170 AO from the same site showing similar remanence components. (*c*) Thermal demagnetization of a specimen 176 AC from the centre of the sill. All directions in this diagram are plotted relative to the present horizontal. Both the stereonet and the X–Y plot show two components: one to the southwest and a second more steeply due west. The southwesterly direction is similar to the tilt-corrected direction of the Nipissing, whereas the due westerly component is similar to the other B directions prior to tilt correction. The former unblocks between 300 and 350°C, and the latter unblocks between 500 and 550°C. (*d*) Thermal demagnetization of a specimen 186 AC from the top of the sill. Two directional components are observed: an A magnetization to the southeast and a B magnetization to the southwest. The intensity decay curve and the X–Y plot suggest that A may reside in two remanence phases. Some A component is unblocked between 300–325°C, but some still remains up to 540°C, at which point a pure B direction with an unblocking temperature of 620°C is found.

The fold test

Two bodies of Nipissing diabase were sampled. Sites H, J, K, and L are in a sill that is presently dipping to the south. Sites A–G are in the limbs of an asymmetrical syncline that is gently plunging to the north. The attitude of the beds at each sampling locality was measured from the nearest sedimentary unit and verified by comparing with the values given on the published geological map of the area (Ontario Department of Mines Map 2017). Tilt corrections were made by simple rotation of the magnetization about the locally measured horizontal strike direction. It is possible that a minor plunge correction should be applied but the effect would be negligible (Ramsay 1967).

As discussed in the previous section a number of distinct remanence phases could be resolved. In general the A magnetizations have lower blocking temperatures than the B magnetizations. Assuming that these remanences were acquired purely by thermal overprinting then the acquisition sequence would be $B \rightarrow A$. Later it will be shown that the sequence $A \rightarrow B$ would lead to a geologically implausible reconstruction.

Within the A remanences, three groups of directions can be identified (Table 1). Sites H and K have a mean *in situ* direction, irrespective of polarity, around $D = 108°$, $I = +31°$ ($\alpha_{95} = 8°$ specimen unit weight). This is not significantly different from the Grenville Front overprint direction of $D = 111°$, $I = +27°$ ($\alpha_{95} = 7°$) reported by Palmer *et al.* (1977) from this same region and there is no reason to suppose that the overprint reported here has a different origin. Sites D and G in the southern limb of the syncline and L in the main Sudbury sill have a mean *in situ* direction of $D = 149°$, $I = +54°$ ($\alpha_{95} = 6°$). Comparison of Fisher precisions before and after tilting indicates that this remanence postdates the folding at the 99% confidence level. Similarly sites A, B, and C in the north limb of the syncline have a mean direction of $D = 103°$, $I = +73°$ ($\alpha_{95} = 6°$); this remanence also appears to postdate the folding with the same degree of confidence. The A magnetizations from each limb of the syncline are, however, directionally quite distinct (Fig. 4*a*). If the A magnetization was acquired synchronously on each limb then there has been relative rotation between the two fold limbs subsequent to the time when

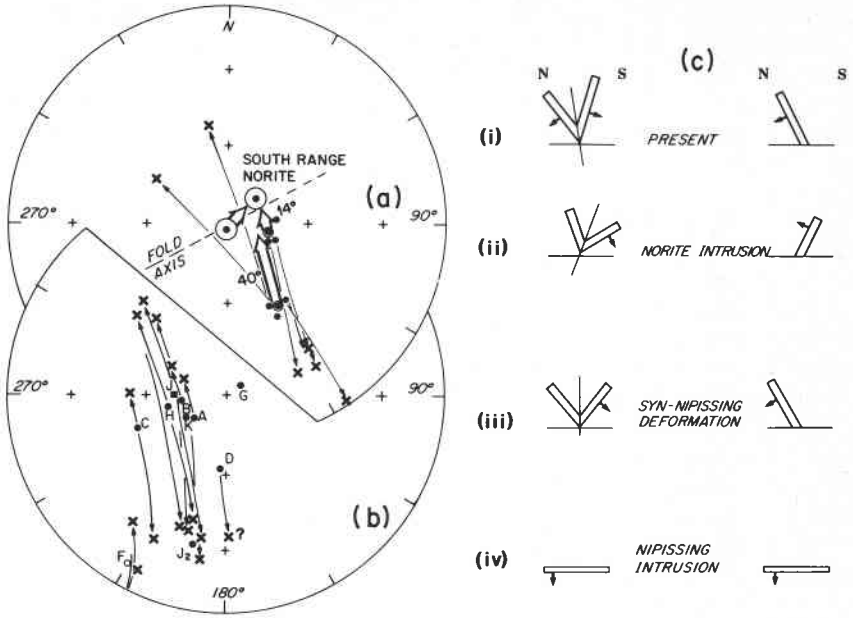


FIG. 4. Paleomagnetic evidence for the tectonic evolution of the two Nipissing sills. (*a*) Equal-angle stereographic projection of the *A* remanence components from each limb of the syncline and from the main sill. Closed (open) dots indicate positively (negatively) inclined remanence vectors. Dots indicate directions prior to tilt correction; crosses are tilt-corrected directions. In both cases these two clusters of directions are better grouped prior to tilt correction. To match the *in situ* directions to the south range norite direction requires rotations of, respectively, 14 and 40° for the north and south limbs of the syncline and 40° for the sill. (*b*) Equal-angle stereographic projection of the *B* remanence components. Plot conventions are the same as for (*a*). To discern the rotations implied by the *B* magnetizations, they must first be corrected for the rotations suggested by the *A* magnetizations. This causes the directions to be rotated first to the north, then (by correcting for the residual dip) to the south. After this second rotation all the directions cluster in the southwest quadrant. (*c*) (*i*) Present geometry of the two sills: one a tight syncline and the other steeply dipping to the south. (*ii*) Configuration of the Nipissing sills at the time of acquisition of the *A* remanence; the syncline is more open, with the axial plane dipping to the north, and the main sill is overturned, dipping and facing north. (*iii*) The agreement between the (*in situ*) J_2, the partially rotated *D*, and the tilt-corrected *A* directions suggests an intermediate post-Nipissing cooling configuration. The main sill has a similar orientation to that of the present day, and the axial plane of the syncline is subvertical. (*iv*) Configuration of the Nipissing sills suggested by the *b* remanence. Both sills were approximately flat-lying; inaccuracies in the data permit this to be a basin and dome structure similar to that presently seen on the Cobalt plate.

A was acquired. Since the total relative rotation (25°) permitted by the data does not allow either limb to be rotated to the present horizontal, the question arises: How much rotation should be apportioned to each limb?

Paleomagnetic investigations of the 1250 Ma Sudbury dike swarm do not reveal any evidence for folding in the Sudbury region (Larochelle 1966; Palmer *et al.* 1977). Therefore the overprint must be older than 1250 Ma. The most likely candidate in the Sudbury region is thermal overprinting by the intrusion of the nearby Sudbury irruptive. The portion of the Sudbury irruptive that is closest to this locality is the noritic phase of the South Range. In Fig. 4*a* the two overprint magnetizations are compared with the initial norite magnetization (Morris 1980), which has been corrected for the same fault-block rotation that the south range has undergone relative to the remainder of the Sudbury irruptive. All three points lie close to the same great circle, but the two overprint magnetizations are on the same side of the corrected norite direction. This indicates that both limbs of the syncline have been rotated in the same sense, the north limb by 14° and the south limb by 40°. After this correction the syncline is more open and the axial planar surface now dips to the north. The sill must also be rotated by 40°, which means that at this point it was slightly overturned (Fig. 4*c*(*ii*)).

To further unravel the fold sequence one must first correct the *B* magnetizations and calculate the residual dips by using the rotations as just defined by the *A* magnetizations (Fig. 4*b*). Applying the residual tilts to the corrected *B* magnetizations produces a clustering in the southwest quadrant. To apply the fold test the normal polarity direction for site F has been adopted. A fold test on the corrected *B* magnetizations of sites A, B, C, F (norm), H, I, J, and K is positive at the 99% probability level (McElhinny 1964). It would appear that the site F

magnetization is crucial to the validity of this test. It is the only site from the north dipping limb of the syncline. All others are from south facing strata. However, there is some difference in the amount of dip between the two south facing strata. Upon repeating the test, but excluding site F, a positive fold test is still found, but the probability is reduced to 95%. As further confirmation of the validity of this test, we note that the mean direction ($D = 199°$, $I = +33°$, $\alpha_{95} = 7°$) is quite similar to other directions reported from the Nipissing diabase (Symons 1970; Symons and Londry 1975; Morris 1979). These tilt corrections suggest that when the Nipissing sills were first intruded they approximated flat-lying sheets. That some tilt corrections appear to be too great by small amounts and the mean inclination is slightly too shallow are compatible with the diabase having been intruded into already gently folded Huronian strata, as seen now on the Cobalt plate. This is one possible explanation for the minor divergence of the normal *F* magnetization; others include polar drift between polarity reversals, and more probably inexact tilt correction, possibly involving rotations about fold axes that were not exactly horizontal or similarly inclined, as assumed here (Macdonald 1980).

Finally, there are three *B* magnetizations (at D, G, J_2) that have not yet been considered. A ready explanation of the *G* magnetization is not apparent. Both D and J_2 are in the more coarse-grained central portions of sills. J_2 as discussed above has low blocking temperatures (<350°C) and has a direction relative to the present horizontal surface that is identical to the totally tilt-corrected directions in sites H and K from the margins of the same sill. J_1, the high-temperature direction at site J, is identical to the site H and K direction. Unless J_2 is a much later overprint it is necessary to find some geological explanation for the identity of the tilt-corrected high-temperature direction and the *in situ* low-temperature direction. Folding of an intrusive sheet that is slowly cooling provides the best explanation. The sites at the margins would be cooled much more quickly than the centre. If syntectonic intrusion were important the high blocking-temperature magnetic fraction at the centre of the sill could record the local geomagnetic field direction prior to any tilting, whereas the lower blocking-temperature fraction could record nearly the same local field direction after folding had been completed. This would result in the high and low blocking-temperature phases at the centre of the sill having different remanence directions that were acquired with approximately the same ambient field direction. From this information it is possible to define the sense and magnitude of rotation the body underwent while cooling.

The agreement between the tilt-corrected *B* directions, the *in situ* J_2 directions, and the accepted Nipissing directions (Morris 1979) indicates that the sill was approximately flat lying when initially intruded. When it finally cooled after undergoing tectonic rotation it then occupied its present-day orientation (Fig. 4*c*(*iii*)). A corollary of this interpretation is that the region between the Sudbury Basin and the Grenville Front was tilted to the south prior to the intrusion of the norite, but after the initial folding of the diabase. During this event the sill was rotated by some 60°. Further evidence for a pre-norite intrusion and post-initial-folding tilting in the Nipissing comes from site D, which also forms the central portion of a sill. Rotating (by approximately 30°) the fold axial plane of the reconstructed syncline (after removal of the post *A* magnetization folding) brings the axial plane back to vertical and brings the D_B direction into closer agreement with the mean Nipissing direction (Fig. 4*c*(*iii*)). This suggests that the central portion of this sill too only cooled below the effective Curie point (here 425°C) after the initial folding, and that it was subsequently tilted prior to the second later post *A* magnetization folding event. It might be noticed that the proposed tilting is greater (60°) near the irruptive margin, and less (30°) further away.

The above interpretation assumes that the remanence acquisition sequence was $B \rightarrow A$. By adopting this sequence it has been possible to construct a tectonic history for these Nipissing sills that fits the geological record; the preferred older direction (the tilt-corrected *B* magnetizations) agrees with a direction (N2) that dates from the original intrusion of the diabase (positive contact test, Morris 1979). With the alternative sequence $A \rightarrow B$ the *A* magnetizations must date from the time of intrusion, yet there is *no* possible way to bring the *A* magnetizations in agreement with either of the two most commonly observed Nipissing directions (N1 at $D = 270°$, $I = 65°$, or N2 at $D = 040°$, $I = -40°$). Hence with this sequence it is necessary to accept that either there is yet another real Nipissing direction (not yet reported in any other studies) or the supposedly older *A* remanence was acquired as an overprint. Neither of these seems likely. This is further emphasized by considering the *B* magnetizations. *In situ* the *B* directions have close similarity to the N1 directions ($D = 270°$, $I = +65°$) found in the Nipissing on the Cobalt plate. This produces the improbable situation where the supposedly older *A* magnetization was an overprint, and the younger *B* magnetization records an intrusive event. A requisite of this is that the *A* magnetization be imposed by a regional metamorphic event, after which the sills were folded and rotated. Following this the *B* magnetization records a subsequent intrusive event that was not followed by tectonism. This scenario cannot be valid for two reasons. Firstly, if the metamorphic event was produced by the norite it requires that the second Nipissing intrusive event (*B*) postdate intrusion of the norite.

Secondly, it requires that these sills exhibit evidence of multiple intrusions. No evidence to support either of these two observations exists. Therefore we can discount the sequence $A \rightarrow B$ and accept the history defined by the sequence $B \rightarrow A$ as outlined above.

Regional tectonic significance

Card (1979) and Brocoum and Dalziel (1974) agree there has been only one tectonic event in the region south of the Sudbury Basin after formation of the Sudbury irruptive. Brocoum and Dalziel (1974) consider this event to have produced all the structures in the Sudbury region. Card (1979) conversely suggests that these structures represent the summation of four distinct tectonic episodes.

From the evidence presented in the previous section it would appear that paleomagnetism appears to favour the second alternative. It was shown that the Nipissing diabase in the region south of the Sudbury Basin shows conclusive evidence of two, and possibly three, tectonic events affecting this region. The oldest tectonic event appears to be related to the initial intrusion of the Nipissing at some time around 2150 Ma. (The exact age of this event is uncertain because it is uncertain as to which of the two Nipissing poles (N1 and N2, Morris 1979) the radiometric age should be attributed.) The next youngest event is a regional tilting of locally variable degree, which possibly records the Sudbury event; radiometric age constraints on this event are 1840–2150 Ma. The youngest tectonic event is a regional folding event that postdates the intrusion of the noritic phase of the Sudbury irruptive. A preferred time interval would be 1700–1840 Ma. This suggests that the Grenville event imposed little or no regional folding on this area.

It is possible to proceed one step further back in time with this regional tectonic analysis. Paleomagnetic fold tests on Nipissing diabase from localities where it outcrops outside the Sudbury region have consistently shown evidence for *no* significant deformation of the Nipissing (Symons 1970, 1971; Symons and Londry 1975), but there is a positive fold test on Gowganda argillites, the unit into which many of the Nipissing diabases are intruded (Morris 1977). Therefore, there is evidence of a fourth tectonic event affecting this region of the Southern Structural Province. Time bounds on this older event are crudely set by the intrusion of the Nipissing at 2150 Ma and the deposition of the Gowganda at approximately 2300 Ma.

The four tectonic episodes recognized paleomagnetically very closely agree with those defined geologically by Card (1979).

Conclusions

Unlike other areas where the Nipissing diabase outcrops in broad basin and arch structures, in the area south of the Sudbury Basin it occupies tightly folded synclines and anticlines. A paleomagnetic study of two sills from this zone (one dipping uniformly to the south, and the other occupying an asymmetrical syncline) suggests that it has undergone a complex tectonic history. Positive fold tests of remanence indicate that there have been two quite distinct folding episodes. The younger event, which postdated the intrusion of the norite (ca. 1850 Ma) but predated the intrusion of the Sudbury dyke swarm (1250 Ma), tightened an initially more open syncline, and rotated the other sill from an overturned position to a uniform southerly dip. The older deformational event predated the intrusion of the norite and must therefore have occurred during the period 1850–2150 Ma. Prior to this event the sills approximated flat-lying sheets. Conclusive evidence exists, therefore, for at least two separate folding events in the Nipissing.

Support also exists for a third event between the two phases outlined above. The equivalence of the low-temperature *in situ* directions from the centre of the sills to the high-temperature directions from the sill margins is compatible with the concept of syntectonic intrusion. This interpretation requires that, upon completion of the first folding event when remanence blocking of the low-temperature components of the B magnetization (J_2, D_B) was finalized, the main Sudbury sill must have had an orientation similar to its present one. To produce this special structural geometry at the time of acquisition of the A magnetization requires that the sill be overturned by some other tectonic event. This indicates there have been *three* deformational events affecting the Nipissing in the Sudbury region. This intermediate event must have occurred prior to the intrusion of the norite. One could speculatively interpret this event as the extrabasinal signature of the so-called Sudbury event, be it meteorite impact or crypto-explosion.

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